Harish Gurrapu | Data Engineer

CA, USA | +1 (510) 513 9509 | harishgurrapu2@gmail.com | LinkedIn

SUMMARY

Data Engineer with 4+ years of experience designing scalable data pipelines and delivering analytics-ready datasets in insurance and enterprise environments. Expert in SQL, Python, Spark, and cloud platforms, with strong experience in ETL development, incremental processing, and data quality validation. Improves reporting accuracy, performance, and decision-making by building curated data models and KPI reporting layers. Partners with analysts and business stakeholders to translate requirements into reliable data solutions that support governance, operational reporting, and performance insights.

TECHNICAL SKILLS

Data Engineering: ETL/ELT Development, Batch & Incremental Processing, Data Modeling, Data Warehousing, Pipeline Performance Optimization, Data Validation & Reconciliation, Monitoring & Failure Handling
Programming & Querying: Advanced SQL (window functions, query tuning), Python, PySpark, Data Transformation & Automation
Cloud & Platforms: AWS (S3, Glue, Redshift), Azure Data Factory, Azure Synapse, ADLS, Cloud Data Lake & Warehouse Architecture
Big Data & Orchestration: Apache Spark, Spark SQL, Apache Airflow, Workflow Scheduling & Dependency Management
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB
Analytics & BI Enablement: Power BI, KPI Metrics & Data Modeling, Dashboard Data Preparation, Trend & Variances Analysis, Exploratory Data Analysis (EDA), Reporting & Data Storytelling
Data Quality, Security & Governance: Data Quality Checks, Schema Validation, RBAC, PII Data Handling, Compliance Awareness
Engineering Practices: Git, CI/CD Exposure, Agile, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Liberty Mutual Insurance, MO, USA | Apr 2025 - Present
Data Engineer

- Built and maintained ETL pipelines processing over one million records daily, increasing data availability and reducing manual preparation effort by **35%** for analytics and reporting teams.
- Implemented incremental loading and validation rules, reducing data mismatches by **30%** while eliminating duplicate and incomplete records to strengthen data reliability for reporting and downstream analytics.
- Optimized complex SQL queries and transformations, cutting dashboard runtimes from minutes to seconds and enhancing performance for high-usage operational and management reports.
- Developed PySpark transformations for large datasets, improving batch stability and lowering pipeline failures by **25%** through optimized partitioning, error handling, and execution logic.
- Orchestrated workflows using Apache Airflow, ensuring timely dataset delivery while enhancing pipeline monitoring, scheduling efficiency, and operational transparency across environments.
- Delivered curated KPI datasets for Power BI dashboards, increasing leadership visibility into performance trends and key business metrics used for planning and strategic evaluation.
- Per formed data profiling and anomaly detection to identify quality issues, strengthening reporting accuracy and increasing stakeholder confidence in business metrics.
- Partnered with analysts and business stakeholders to translate requirements into analytics-ready datasets and reporting models, reducing rework and accelerating delivery of actionable insights.

Mphasis, India | Sep 2019 - Dec 2022
Data Engineer

- Developed batch pipelines ingesting relational databases, APIs, and flat files, ensuring consistent data availability for finance and operations reporting across global business units.
- Automated ingestion and transformation workflows using Python, reducing manual processing effort by 40% while improving consistency, reliability, and processing efficiency.
- Built SQL transformation layers and reporting marts that enhanced reporting accuracy and enabled faster business insights across operational and financial datasets.
- Processed large datasets using Apache Spark, reducing batch runtimes by 20% through partition optimization and execution tuning for high-volume workloads.
- Implemented validation and reconciliation checks, increasing trust in reporting outputs and reducing downstream correction cycles.
- Delivered reporting-ready datasets enabling KPI tracking, variance analysis, and performance monitoring across departments.
- Supported ad hoc analysis requests for planning and forecasting, enabling teams to access timely insights and respond quickly to operational challenges.
- Maintained pipeline version control and structured deployments using Git, improving release stability, reducing production defects, and strengthening change management practices.

EDUCATION

Masters in Computer/information technology services administration and management | Aug 2025
Avila University, MO, USA
Bachelors in Bachelor of Science (B.Sc.) in Statistics, Mathematics & Computer Science | Jul 2019
Satavahana University, India